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A Lithium Salt of a Lewis Acid-Base Complex of Imidazolide for Lithium-Ion Batteries

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We report the synthesis of a class of lithium salts in which the two strongly Lewis basic sites of lithium imidazolide are each complexed by the Lewis acid BF_3 to yield lithium bis(trifluoroborane)imidazolide [$LiIm(BF_3)_2$]. The anion has a high degree of charge delocalization over the aromatic ring and two four-coordinate boron centers and has only very weakly basic sites on the periphery which keep the coordinating ability to the lithium cation low. The salt has a conductivity of 5.06 mS/cm in 1:3 ethylene carbonate:ethylmethyl carbonate solution and has high oxidative stability. Lithium cells made with common anode and cathode materials using an electrolyte containing this salt had performance similar to cells using an electrolyte containing $LiPF_6$.

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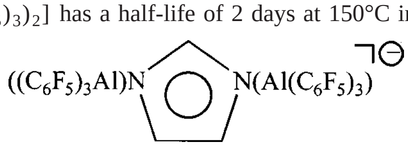
The major factors that are considered in choosing an electrolyte for a consumer battery are cost, performance, and safety. Much of the effort toward developing an electrolyte for lithium and lithium-ion (Li-ion) batteries has focused on the synthesis of new lithium salts; however, none have met all three requirements adequately. In terms of performance an electrolyte must have high conductivity, high thermal stability, and electrochemical stability. Most salts for Li-ion batteries can be classified as belonging to one of three families of salts. Anions with a central atom in its highest oxidation state surrounded by anionic ligands form the first class of salts. Some examples include PF_6^-, BF_4^-, bis(oxalato)borate,[1,2] and other chelatoborates.[3,4] Lithium salts of anions with an atom in its most reduced state that is bonded to one or more highly electronegative groups form a second class of conductive salts. Examples include lithium salts of trifluoromethanesulfonate, perfluorinated carboxylates and amides,[5,6] or methides[7] with highly electronegative groups such as fluorinated sulfonyl groups. A third and less common class of lithium salts include those where the anion is a caged complex as in $B_{10}Cl_{10}^{2-}$.[8]

Recently, a new family of weakly coordinating anions has been developed for use in olefin polymerization catalysts by LaPointe *et al.*,[9] an example of which is $Im(Al(C_6F_5)_3)_2^-$ (Im = imidazole, $C_3N_2H_3$), shown below. In this family of salts, a monoanionic species with multiple Lewis basic sites is complexed by multiple Lewis acids. By forming a Lewis acid-base complex at each basic site of the parent anion, the Lewis basicity is dramatically reduced thus diminishing the affinity of the anion for the cation, and reducing ion-pairing. The olefin polymerization catalysis investigated by LaPointe *et al.*, requires that the anion be dissociated from cation in an organic solvent with a low dielectric constant such as toluene. The dissociation of the anion from the cation (*i.e.*, low degree of ion-pairing) is also important for achieving a highly conductive lithium salt for Li-ion batteries. The addition of a Lewis acid to a low conductivity electrolyte solution containing a salt such as LiF, CF_3CO_2Li, and $CF_3CF_2CO_2Li$ was previously shown by Sun *et al.*, to improve the conductivity,[10-12] presumably by forming a complex with the anion. Lithium salts similar to the ones prepared by LaPointe *et al.*, therefore should be highly conductive, and because they incorporate the addition of multiple Lewis acids, which act as strong electron withdrawing groups to the anion, they should have high oxidative stability. The thermal stability of these salts was reported to be high; for example, [$HN(C_{18}H_{37})(CH_3)$][$Im(Al(C_6F_5)_3)_2$] has a half-life of 2 days at 150°C in benzene[9]



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Yardney Technical Products recently provided a preliminary report on the key performance parameters of several salts falling into this family of lithium salts.[13] This paper reports the synthesis of the most promising of these compounds, namely, lithium bis(trifluoroborane)imidazolide ($LiIm(BF_3)_2$), and more fully describes its electrochemical stability and conductivity data. In addition, we show that cells with this electrolyte have comparable performance to those containing $LiPF_6$.

Experimental

All preparations and physical measurements were carried out with rigorous exclusion of air and water. Schlenk and glove box techniques were employed with purified argon used as an inert gas when required. All reagents and solvents were reagent grade or better. Imidazole was purchased from Aldrich and used as received. Boron trifluoride diethyl etherate and *n*-butyl lithium were both purchased from Alfa Aesar and used as received. The following solvents were dried by distillation from the indicated drying agent: dichloromethane (P_2O_5), toluene (Na), and d_6-acetone (4 Å molecular sieves). Battery grade ethylmethyl carbonate (EMC), ethylene carbonate (EC), diethyl carbonate (DEC), and dimethyl carbonate (DMC) were purchased from EM Science and used as received.

Physical characterization.—NMR spectra were recorded using a Bruker AC 250 or a JEOL GSX 400 MHz NMR spectrometer. Chemical shifts (δ) are reported relative to $Si(CH_3)_4$ ($\delta = 0$ for ^1H NMR) and $CFCl_3$ ($\delta = 0$ for ^{19}F NMR). Negative- and positive-ion electrospray mass spectra were performed on a Micromass Quattro II mass spectrometer with cone voltages ranging from 15 to 70 V. Ten μL of a sample (100 μg/mL in acetone) were injected into a Rheodyne injector with an acetonitrile flow.

Synthesis of lithium imidazolide (LiIm).—A slurry of imidazole (5.00 g, 73.5 mmol) in toluene (50 mL) at room temperature was treated with 28 mL of a 2.65 M *n*-BuLi (74.2 mmol) solution in hexanes. This solution mixture was then refluxed for 3 days during which time the slurry acquired an off-white color. The slurry was then filtered over a medium glass frit and the solid was washed with two 10 mL portions of toluene and then dried under vacuum to yield an off-white powder. Yield: 5.40 g, 99.4%.

Synthesis of $LiIm(BF_3)_2$.—A slurry of LiIm (5.00 g, 67.6 mmol) in CH_2Cl_2 (100 mL) was treated with $BF_3(Et_2O)$ (19.6 mL, 154 mmol) and the mixture was refluxed for 5 days during which time the slurry became yellow. The solid was then dried under vacuum to yield an off-white solid. Yield: 13.77 g, 97.1%. The solid was then dissolved in a minimal amount of DEC and filtered. A small amount (0.23 g, 1.6% of expected yield) of an insoluble, unidentified solid was left on the filter and was discarded. To precipitate the salt as a solid from the filtrate so that it could be physically characterized,

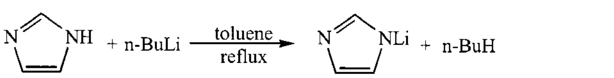
CH_2Cl_2 was added. This precipitated solid was then collected and dried under vacuum at 60°C. Yield: 8.63 g, 61%. ^1H NMR (acetone-d_6) δ 7.87 (s, 1H), 7.08 (s, 2H); ^{19}F NMR (acetone-d_6) δ−147.5 (q, $J_{B\text{-}F} = 13$ Hz).

Electrochemical characterization.—Conductivities of electrolyte solutions at varying temperatures and concentrations of the lithium salts in a 1/3 EC/EMC mixture were measured using a Metrohm 712 conductivity meter. The cell assembly used an Orion 018010 or a Metrohm 712 conductivity cell, both of which have platinized platinum electrodes with cell constants of about 1 cm^{-1}. Cells are filled and sealed inside a glove box under an argon atmosphere. The measurement temperatures were controlled to within 1°C using a Tenney Environmental temperature chamber. Cyclic voltametry was performed on a Solartron SI 1287 potentiostat with a K0264 MicroCell Kit using a three-electrode cell with a platinum working electrode (0.0397 cm^2), and lithium counter and reference electrodes.

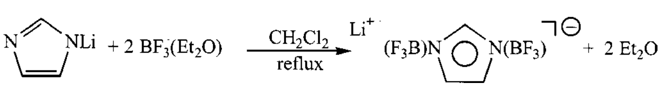
Coin cell measurements.—The compatibility of the electrolyte solution with common anode (mesocarbon microporous bead (MCMB) carbon) and cathode ($LiNi_{0.8}Co_{0.2}O_2$) materials was tested by cycling cells prepared with lithium metal as the anode in 2325 size coin cells in a standard button cell configuration. The $Li/LiNi_{0.8}Co_{0.2}O_2$ and Li/MCMB cells were never charged faster than 0.4 mA/cm^2 to slow the formation of Li dendrites during the deposition of lithium. The cells were cycled using an Arbin battery cycler. The $Li/LiNi_{0.8}Co_{0.2}O_2$ cells were formed using a C/20 charge and discharge cycle followed by two C/10 cycles. The Li/MCMB cells were formed at a C/7 rate for the discharge and charge.

Results and Discussion

$LiIm(BF_3)_2$ was prepared in two steps as shown in Eq. 1 and 2. Both reactions proceed in near quantitative yield. To remove the diethylether, which is not easily removed under vacuum, the salt was dissolved in DEC and then precipitated with the addition of CH_2Cl_2. The possibility of using BF_3 gas in place of an adduct of BF_3 is currently being investigated and should eliminate the need for this step



[1]



$LiIm(BF_3)_2$ [2]

The $LiIm(BF_3)_2$ was characterized by NMR spectroscopy and mass spectrometry. The ^1H and ^{19}F NMR spectra of $LiIm(BF_3)_2$ are shown in Fig. 1 and 2, respectively. Important features to note are that the major signals are consistent with expected product and the relative purity of the compound. The small impurity (∼5%) observed in the ^{19}F NMR (Fig. 2) spectrum is $LiBF_4$, which is a conductive and electrochemically stable salt used in Li-ion systems. This signal was confirmed to be $LiBF_4$ by spiking a NMR sample with an authentic sample and by mass spectrometry. Negative-ion electrospray mass spectrometry, a very "soft" technique that causes little fragmentation, was also used to characterize the new salt. This technique is especially useful for materials that are already ionized in solution, and typically gives the mass of the parent anion as well as some fragmentation ions. The mass spectrum has a peak at 203 mu (Da/electron) with an isotope pattern that is consistent with the singly charged, parent $Im(BF_3)_2^-$ anion and another peak at 135 mu which corresponds to the mass of the anion after loss of one BF_3 molecule. One additional peak at 87 mu was observed which was assigned to BF_4^- since this was also observed in the ^{19}F NMR

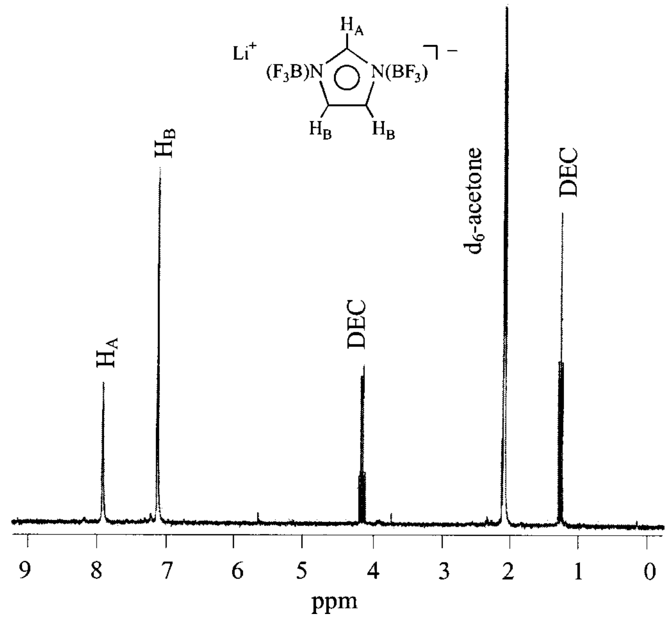


Figure 1. A 250 MHz ^1H NMR spectrum of $LiIm(BF_3)_2$ in d_6-acetone.

spectrum. The mechanism for the formation of $LiBF_4$ and the thermal stability of this new salt is still under investigation and will be reported in the future.

Conductivity.—The new salt is very soluble and conductive over a wide temperature range. Figure 3 shows the conductivity of a series of solutions with concentrations ranging from 0.25 to 2 M in 1/3 EC/EMC from −50 to +80°C. No precipitation occurred in solutions with concentrations as high as 2 M in 1/3 EC/EMC at temperatures as low as −50°C. Solutions having $LiIm(BF_3)_2$ concentrations of 0.75 to 1 M are most conductive near ambient temperatures as solutions with high concentrations tend to be viscous and those with low concentrations have a limited number of charge carriers. The conductivity of a 1 M solution of $LiIm(BF_3)_2$ (5.06 mS/cm) is more than twice as conductive than a 1 M $LiBF_4$ (1.78 mS/cm) in 1/3 EC/EMC at 20°C which shows that the $LiBF_4$

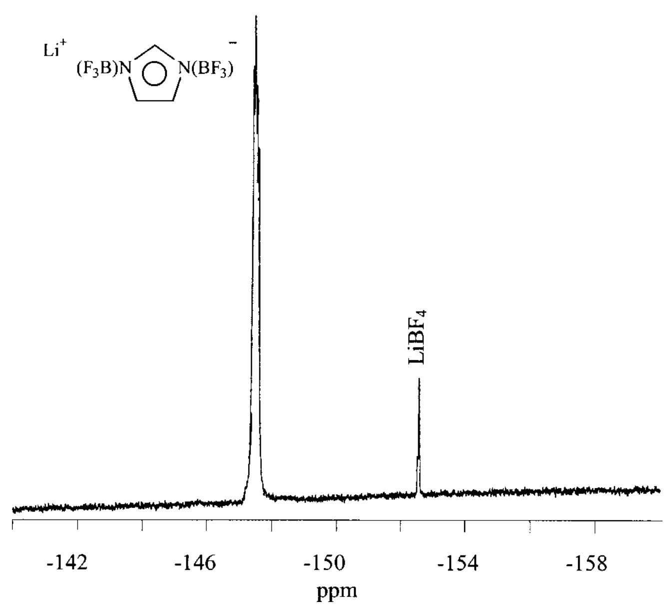


Figure 2. A 235 MHz ^{19}F NMR spectrum of $LiIm(BF_3)_2$ in d_6-acetone.

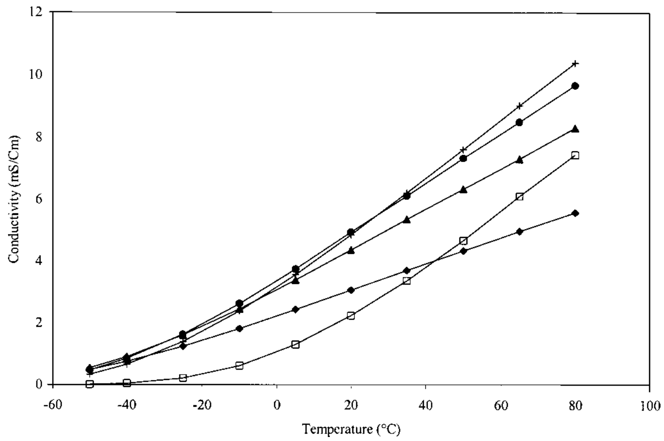


Figure 3. Conductivity of LiIm$(BF_3)_2$ solutions in 1/3 EC/EMC from −50 to +80°C at various concentrations (♦: 0.25 M, ▲: 0.50 M, ●: 0.75 M, +: 1.0 M, □: 2.0 M).

impurity is not responsible for the observed high conductivity. For comparison, a 1 M $LiPF_6$ in 1/3 EC/EMC at 20°C is 7.61 mS/cm.

The structure of the anion is ideal for yielding a highly conducting solution when it is dissolved in a polar solvent. First, the salt has a molecular weight of only 209.6 which helps keep solution viscosity low at high concentrations. Second, the structure of the anion allows the negative charge to be dispersed over the entire anion. The anionic five-membered imidazole ring is aromatic allowing the negative charge to be dispersed over the ring. Second, each of the BF_3 groups form a bond to a different nitrogen atom on the imidazole ring thus creating two four-coordinate boron centers, each of which have an equal amount of partial negative charge due to the symmetry of the anion. Furthermore, there are no strongly basic groups present on the periphery of the anion to which the cation could bind strongly which allows the solvent to more effectively compete for coordination sites of the lithium cation resulting in greater solvation and less ion-paring. The overall effect is a low solution viscosity and a low degree of ion-pairing, both of which contribute to high ionic conductivity.

Oxidative stability.—The high oxidative stability of the new salt is illustrated by the cyclic voltametry data in Fig. 4. The voltammograms of 1 M LiIm$(BF_3)_2$ and $LiPF_6$ solutions in DMC solution were obtained at a sweep rate of 10 mV/s between 3 and 6 V with a platinum working electrode. The electrolyte solution containing LiIm$(BF_3)_2$ is electrochemically stable to 4.85 V *vs.* Li, about 0.5 V below that of a $LiPF_6$ solution, but still well above the voltage range of our Li-ion cell (4.1 V). The highly electron withdrawing nature of the multiple Lewis acidic BF_3 and the aromaticity of the anion contribute to the high oxidative stability of this anion.

Table I. Comparison of several performance parameters for $LiNi_{0.8}Co_{0.2}O_2$ cells containing LiIm$(BF_3)_2$ and $LiPF_6$.

	LiIm$(BF_3)_2$	$LiPF_6$
Irreversible capacity	10.3	10.0
Capacity at C/7 rate (mAh/g)	171.8	170.9
Capacity at C/1.75 rate (mAh/g)	157.6	155.0
Capacity at 2.25 C rate (mAh/g)	129.7	119.5
Average coulombic efficiency (%)	98	99

Compatibility with electrode materials.—The compatibility and performance of the electrolyte containing the new lithium salt with these common Li-ion cell anode and cathode materials have been demonstrated in coin cells. These cells had lithium anodes and either MCMB carbon or $LiNi_{0.8}Co_{0.2}O_2$ cathodes.

The cells containing the cathode material $LiNi_{0.8}Co_{0.2}O_2$ had an overall performance similar to that of identical cells prepared with $LiPF_6$ as shown in Table I. The irreversible capacity for cells containing LiIm$(BF_3)_2$ and $LiPF_6$ is similar. The overall capacity also compares well to that for a cell with $LiPF_6$ solution in 1/3 EC/EMC as shown in Fig. 5 at all discharge rates up to the 2 C rate, which was the highest rate tested. The Li/$LiNi_{0.8}Co_{0.2}O_2$ cell with LiIm$(BF_3)_2$ lost only 2% of its discharge capacity between cycle 4 and cycle 20 which suggests good cycle life and compatibility with the cathode electrode material.

Generally, the Li/MCMB cells with the LiIm$(BF_3)_2$ salt took longer to utilize the full capacity of the anode material but after about 10 cycles the capacities obtained were close to that of cells containing $LiPF_6$ as illustrated in Fig. 6[13] where the discharge capacities of cells containing LiIm$(BF_3)_2$ and $LiPF_6$ are compared. The irreversible capacity of the LiIm$(BF_3)_2$ and $LiPF_6$ cells was 10.2 and 6.2%, respectively. However, in terms of total capacity lost during the first cycle, the cell containing LiIm$(BF_3)_2$ lost only 0.057 mAh more capacity than the $LiPF_6$ cell. This difference is small (1.2%) in comparison to the discharge capacity of 4.65 mAh for the first cycle of the $LiPF_6$ cell. The coulombic efficiency of subsequent cycles is over 99%, which suggests that the solid electrolyte interface (SEI) layer is fully formed during the first cycle. Furthermore,

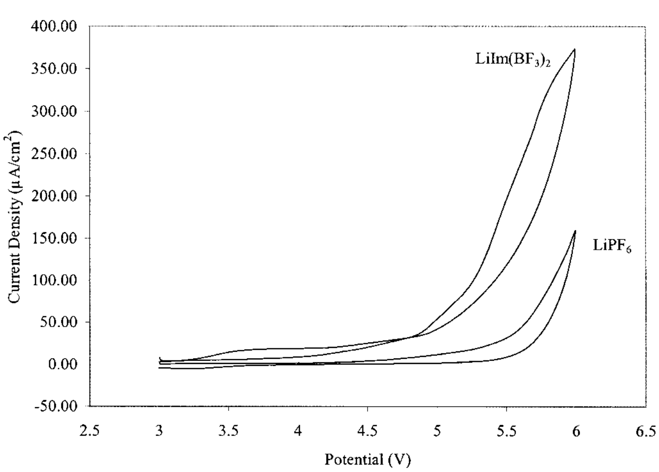


Figure 4. Cyclic voltammograms of 1 M LiIm$(BF_3)_2$ and 1 M $LiPF_6$ solutions in DMC. A sweep rate of 10mV/s was used from 3 to 6 V.

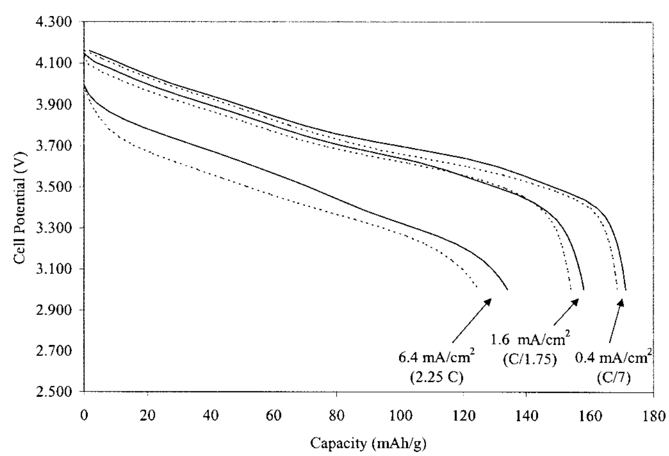


Figure 5. Discharge profile for Li/$LiNi_{0.2}Co_{0.8}O_2$ cell at various rates with (solid line) 1 M LiIm$(BF_3)_2$ in 1/3 EC/ EMC and (dashed line) 1 M $LiPF_6$ in 1/3 EC/EMC.

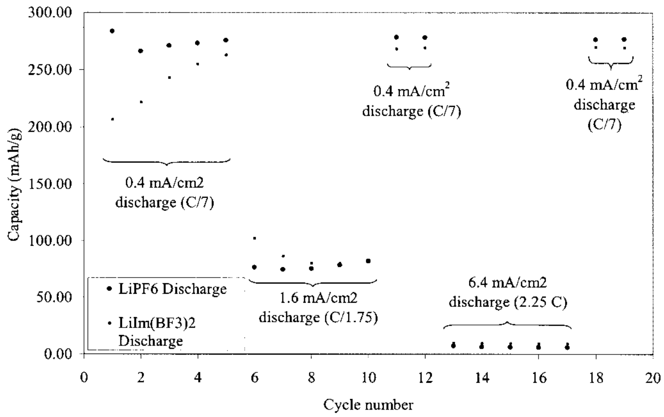


Figure 6. A comparison of the discharge capacities of Li/MCMB cells with 1 M $LiIm(BF_3)_2$ in 1/3 EC/EMC and 1 M $LiPF_6$ in 1/3 EC/EMC. The discharge rate was varied as shown while the cell was charged at a 0.4 mA/cm^2 rate for all cycles.

because the total capacity used during the first cycle is similar to that of a $LiPF_6$ cell, the SEI layer formed with this new electrolyte may be similar to that formed with an electrolyte containing $LiPF_6$.

At high rates of discharge (*i.e.*, intercalation of lithium into carbon which occurs during the charge in a Li-ion cell) the capacity for both Li/MCMB cells is very low as shown in Fig. 6. Because capacities at higher rates are about the same but the conductivities are different for $LiIm(BF_3)_2$ and $LiPF_6$ solutions, the rate capability in these cells is limited by the rate of lithium intercalation into the carbon and not the conductivity of the electrolyte. With such a limited number of cycles and with the capacity increasing in the first several cycles, the only conclusion that can be made regarding fade rate in this cell is that it is very low and appears to offer similar performance to a $LiPF_6$ cell.

Conclusions

A new class of Li salts has been developed for Li-ion batteries in which a central anion with multiple Lewis basic sites is fully complexed by Lewis acids. A lithium salt of a Lewis acid-base complex of imidazolide has been prepared which has high solubility, conductivity, and electrochemical stability at ambient temperatures. The salt is prepared from inexpensive materials in high yield, which may offer a low-cost alternative to $LiPF_6$. The salt is highly soluble even at low temperatures. The ambient temperature conductivity was ~5 mS/cm in a 1/3 EC/EMC solution. The electrochemical stability of an electrolyte solution with this salt is very high (>4.8 V *vs.* Li). An electrolyte containing this new salt performs as well as an electrolyte containing the widely used $LiPF_6$ salt in $Li/LiNi_{0.8}Co_{0.2}O_2$ cells and exhibits low capacity fade with cycling. While it took longer for the full capacity of Li/MCMB cells to be utilized when the new salt was used in these cells, comparable performance to cells containing $LiPF_6$ was obtained after about 10 cycles.

Acknowledgments

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